

PUNJAB FLOOD PROTECTION EMBANKMENTS

Sector: Agriculture

Hazard addressed: Floods

Targeted assets: Buffalo, Cattle, Goat, Sheep, Backyard Poultry, Rangeland



Description: This measure involves construction and strengthening of flood protection embankments along vulnerable riverbanks and livestock production areas to reduce flood damage to livestock systems and associated infrastructure. The intervention protects livestock shelters, grazing lands, and rural settlements from inundation and erosion while enhancing resilience of livestock-dependent communities against extreme flood events.

Priority Districts: Bahawalnagar, Bahawalpur, Dera Ghazi Khan, Jhang, Khanewal, Lodhran, Multan, Muzaffargarh, Rahim Yar Khan, Rajanpur, Toba Tek Singh, Vehari, and Sialkot.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Kilometer (km) of flood protection embankment.

Scale assumptions: Applicable in flood-prone livestock production areas and riverine zones vulnerable to seasonal flooding and riverbank erosion.

Key design parameters: Embankment height and width, stone pitching, toe protection, slope stabilization, drainage outlets, and hydraulic design based on expected flood return periods and river flow conditions.

Time horizon: Start construction: 2030; ready by 2040, continue with periodic maintenance and reinforcement until 2050.

IMPACT

Risk reduction level: High.

Time to effectiveness: Immediate to short-term following completion of embankment construction.

Expected impact: Reduced flood inundation and erosion of livestock areas, protection of livestock shelters and grazing lands, reduced livestock losses, and enhanced resilience of livestock systems against flood hazards.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Hydrological assessments, engineering design expertise, financing support, land availability, and institutional coordination.

Required institutions: Punjab Irrigation Department, Livestock Department Punjab, PDMA Punjab, Local Government Department, and development partners.

Barriers: High capital costs, land acquisition constraints, sedimentation, weak maintenance systems, and potential structural failure during severe flood events if maintenance is inadequate.

COST ASSUMPTIONS

Unit cost: PKR 173 million per km (~USD 617,857 per km).

Capital Expenditure (CAPEX): USD 36.9 million per year (2030 -2040). Includes embankment construction, stone pitching, toe protection, earthworks, drainage structures, and associated civil works.

Operational Expenditure (OPEX): USD 4.0 million per year (2041-2050). Includes inspection, repair works, desilting, vegetation clearing, and reinforcement maintenance.

CO-BENEFITS & TRADE-OFFS

Environmental: Reduced soil erosion and stabilization of vulnerable riverbanks and grazing areas.

Social: Improved livelihood security and reduced disaster risks for livestock-dependent communities.

Economic: Reduced economic losses from livestock mortality, infrastructure damage, and flood-related disruptions.

Potential risks: High upfront investment requirements, possible disruption of natural drainage patterns, and maintenance challenges during extreme flood events.

BENEFICIARIES

Primary beneficiaries: Livestock farmers, pastoral communities, rural households, irrigation departments, and local governments.

Distributional aspects: Particularly beneficial for vulnerable livestock-dependent communities located in flood-prone river corridors and low-lying rural areas.